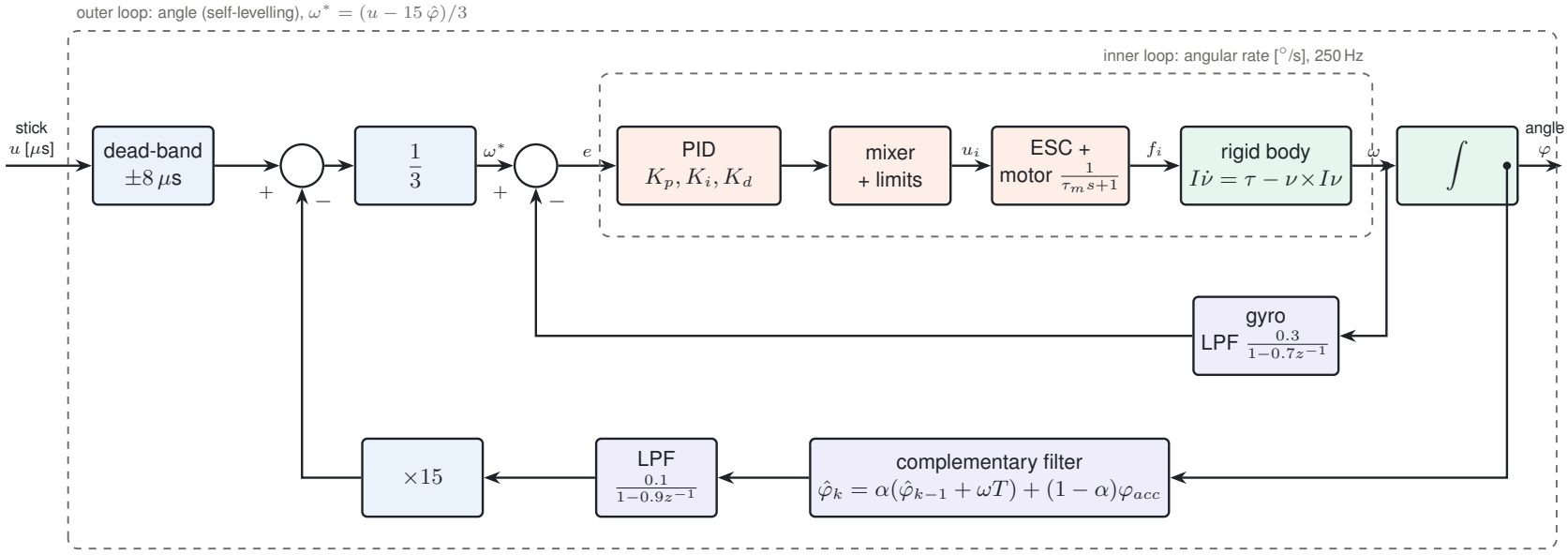

outer loop: angle (self-levelling), $\omega^* = (u - 15\,\hat{\varphi})/3$
inner loop: angular rate [°/s], 250 Hz
stick
u [μs]
dead-band
$\pm 8\,\mu$s
+
−
$\frac{1}{3}$
ω^*
+
−
e
PID
K_p, K_i, K_d
mixer
+ limits
u_i
ESC +
motor $\frac{1}{\tau_m s+1}$
f_i
rigid body
$I\dot{\nu} = \tau - \nu \times I\nu$
ω
$\int$
angle
φ
gyro
LPF $\frac{0.3}{1-0.7z^{-1}}$
$\times 15$
LPF
$\frac{0.1}{1-0.9z^{-1}}$
complementary filter
$\hat{\varphi}_k = \alpha(\hat{\varphi}_{k-1} + \omega T) + (1-\alpha)\varphi_{acc}$